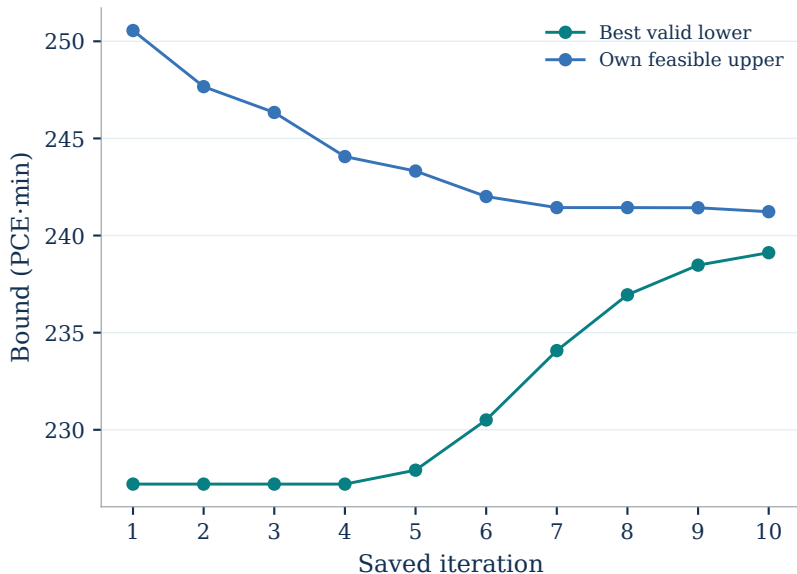


Lagrangian bounds and certified gap

a Hybrid original-cost stage



b Same-instance bound certificate

